

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Same calendar shock, very different failures

Eight SALT tasks share the COVID-era split; their coverage losses vary sharply.

Task	Validation	Post-shift	Drop
Sales payment terms	90.8%	13.7%	77.1 pp
Sales shipping condition	93.5%	21.8%	71.6 pp
Item plant	92.0%	81.4%	10.6 pp
Sales office	99.9%	99.9%	0.1 pp

- Non-randomized APS targets 90% coverage; values are 50-seed means.
- Drop = validation coverage minus post-shift coverage; rounded values may not subtract exactly.

Measure where the model places its reliance

Top-1 concentration is the largest share of total mean absolute SHAP importance.

$$C = \text{largest mean } |SHAP| / \text{total mean } |SHAP|$$

- Compute SHAP on validation data before deployment.
- A large C means one feature dominates the measured attribution.
- C measures reliance concentration; it does not measure how much future data will shift.

Why concentration can amplify vulnerability

The theory explains a mechanism under stated assumptions.

Idealized mechanism

The class-probability model combines a dominant-feature component and a residual component. Shift damages the dominant component while residual behavior stays exchangeable.

Formal conclusion

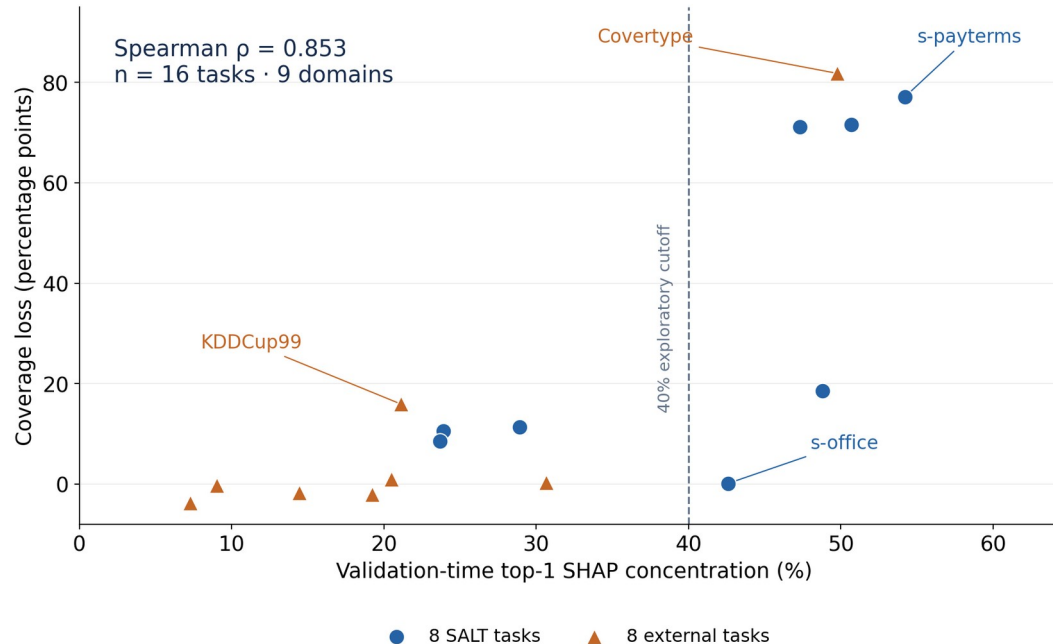
Under the theorem's conditions, a lower bound on expected APS score increases with concentration; a corresponding coverage upper bound worsens.

Empirical boundary

Probability-space additivity is an idealization. Measured TreeSHAP concentration is a proxy, not a guaranteed mixing coefficient.

A strong association across the study tasks

Primary pooled endpoint: Spearman $\rho = 0.853$ across 16 multiclass tasks in 9 domains.



- Reported $p < 0.001$; bootstrap 95% interval $[0.50, 0.96]$.
- The pool includes the original 8 SALT tasks and 8 external tasks.

Exact source data replotted: 8 SALT and 8 external multiclass tasks. The 40% cutoff is exploratory.

Detecting shift and predicting damage differ

On the eight SALT tasks, model-specific reliance is more associated with coverage loss.

Diagnostic	Spearman ρ	When available
Top-1 SHAP concentration	0.833	Before deployment
Native gain concentration	0.667	Before deployment
Across-seed disagreement	0.452	Before deployment
MMD / C2ST / PSI	≤ 0.19	Requires shifted data

- Different information requirements: this is not a same-input forecasting contest.
- Association does not establish a causal effect of concentration.

The 40% rule has useful exceptions

An exploratory flag is not a safe-deployment certificate.

Task	C	Drop	Interpretation
Sales payment terms	54.2%	77.1 pp	High-C failure
Sales office	42.6%	0.1 pp	False positive
Covertime	49.8%	81.8 pp	External failure
KDDCup99	21.1%	15.9 pp	False negative

- At-risk means mean coverage loss > 15 percentage points.
- A protective-feature rule came from one case; it requires further validation.

Scope changes with the classifier

The strongest evidence concerns gradient-boosted multiclass APS predictors.

LightGBM

Within SALT, concentration-drop $\rho = 0.833$. Single-feature reliance aligns with the studied failure mechanism.

Random forest

Within SALT, $\rho = 0.30$. Concentration is a weaker diagnostic in the reported architecture check.

MLP

Within SALT, $\rho = 0.43$. Large losses also occur with low concentration; the top-feature signal misses these cases.

- These are observed model-family differences, not proof of their training causes.

Recovering coverage can reduce usefulness

The diagnostic identifies vulnerability; mitigation remains mechanism-dependent.

Adaptive calibration

ACI increases coverage on severe tasks, but may return very large label sets. Coverage and informativeness must be reported together.

Regularized scores

RAPS reduces loss for some high-class-count tasks, yet does not uniformly repair concentrated-dependence failures.

Retraining

Quarterly retraining offers suggestive partial recovery; the highlighted single-seed effect is not significant after Holm correction.

Model reliance and future coverage

Concentration is a warning signal to test, not a new coverage guarantee.

- Measure validation-time top-1 SHAP concentration for the fitted model.
- Check whether a plausible shift threatens its dominant feature.
- Treat the 40% cutoff and protective-feature rule as exploratory.
- Validate prospectively and monitor both coverage and set usefulness.

Source: github.com/ChorokLeeDev/conformal-covid-uai2026